

# The Perturbation Catalogue: An Integrated Resource for Exploring the Impact of Genetic Perturbations

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ONE SEARCH • THREE MODALITIES • COMPARABLE EVIDENCE

## THE PROBLEM

**A gene change is one question. The evidence is fragmented.**

Different assays, formats, identifiers and processing choices make studies hard to find together—and harder to compare.

### Perturb-Seq

Transcriptome-wide response  
Single cells • DEA • GSEA

### CRISPR screens

Gene-level fitness and phenotype  
Loss/gain of function

### MAVE

Variant-level functional effects  
Deep mutational scanning

## One curated evidence layer

Search by **target** or **dataset**; understand the experimental context without reconstructing every study first.

ONTOLOGY-ALIGNED METADATA

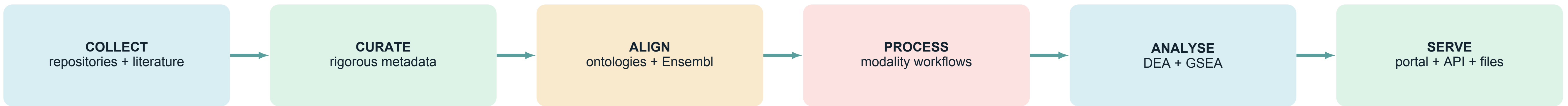
ENSEMBL IDS

VISIBLE PROVENANCE

COMMON ACCESS

## HOW IT WORKS

**From heterogeneous studies to evidence you can use.**



A unified, extensible metadata schema captures study, perturbation, assay, model-system and target context. We reuse established ontologies—and contribute improvements back.

## LIVE NOW

**1,237**  
DATASETS

**19,235**  
HUMAN TARGETS

**3**  
MODALITIES

**5**  
PERTURB-SEQ DATASETS REPROCESSED

1,201 CRISPR screens • 26 Perturb-Seq datasets • 10 MAVE datasets • 36 tissues • 34 cell types • 1,200 cell lines • 169 diseases

Production catalogue snapshot, 24 August 2026.

## UNIFORM REPROCESSING

**Five Perturb-Seq datasets.  
One processing path.**

Reprocessing from raw FASTQ reduces avoidable study-to-study differences and makes analytical provenance explicit.



**LIVE:** Nadig 2025 (Jurkat, HepG2) + Replogle 2022 (K562 essential, K562 genome-wide, RPE1 essential).

Standardised DEA + Hallmark GSEA are searchable beside the curated experimental context.

## EVIDENCE IN CONTEXT

**Search once. Traverse modalities.**

Find a target by symbol, synonym or Ensembl ID; see evidence across Perturb-Seq, CRISPR and MAVE.

### Browse Targets

Explore all gene targets across perturbation experiments. Use the search bar to find genes by symbol, Ensembl ID, or name, and the facets to narrow down results.

☐ CC BY 4.0 (32)

☐ CC BY-NC-SA 4.0 (31)

☐ MIT License (31)

☐ free to use license (9)

☐ CRISPR screen (32)

☐ Perturb-seq (31)

Filter by Tissues

| TARGET NAME                      | PERTURB-SEQ DATASETS                                           | CRISPR-SCREEN DATASETS            | MAVE DATASETS | TOP GSEA TERMS                                                                                                                    | DATA MODALITIES              |
|----------------------------------|----------------------------------------------------------------|-----------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>BRCA2</b><br>ENSG000000139618 | Datasets: 7<br>(Sig. genes up : 127;<br>Sig. genes down : 322) | Datasets: 1,191<br>Sig. hits: 640 | 0             | HALLMARK_MYC_TARGET...<br>HALLMARK_E2F_TARGETS...<br>HALLMARK_OXIDATIVE_P...<br>HALLMARK_G2M_CHECKP...<br>HALLMARK_MITOTIC_SPL... | CRISPR screen<br>Perturb-seq |
| <b>BRCA1</b><br>ENSG000000012048 | Datasets: 7<br>(Sig. genes up : 23;<br>Sig. genes down : 115)  | Datasets: 1,191<br>Sig. hits: 509 | 0             | HALLMARK_MYC_TARGET...<br>HALLMARK_E2F_TARGETS...<br>HALLMARK_OXIDATIVE_P...<br>HALLMARK_OXIDATIVE_P...                           | Perturb-seq<br>CRISPR screen |

## TAKE THE DATA WITH YOU

### PORTAL

Browse targets + datasets

### REST API

Everything shown on the site

### FULL DATA

Parquet + compressed CSV

### METADATA

Downloadable JSON

## NEXT

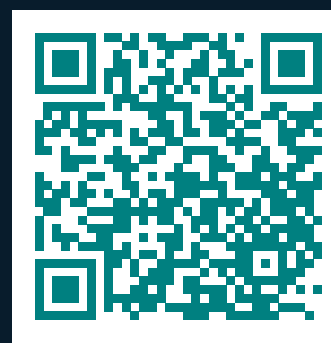
**A living resource, not a one-off integration.**

AI-assisted metadata curation with human review  
Broader modality-specific reanalysis and validation  
Periodic, versioned data releases with transparent provenance

## SHAPE WHAT COMES NEXT

**Bring your hard-to-integrate evidence.**

Suggest datasets or features—or fund a focused, multi-year collaboration around the modalities and questions that matter to your programme.



EXPLORE



CONTACT

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